Library Database Requirements

Library Database Requirements Document

COSC 3380 Database Systems — Group 6 — Checkpoint #2 (revision v3: catalog expanded, subtype integrity enforced)

Revision notes (v3 vs v2)

Changes agreed by the group since v2. Everything marked **(v3)** below is new or changed.

1. **Strict subtype integrity.** Every subtype row must belong to a CatalogItem of the matching type. Each subtype table now carries its own `Item_type` (fixed to its own value) and links to CatalogItem on `(Item_id, Item_type)` together.
2. **Two new categories: Newspaper and Research Paper.** They circulate like any other item. New tables: `Newspaper`, `ResearchPaper`, and the join table `PaperAuthor`. `Item_type` gains `'newspaper'` and `'researchpaper'`.
3. **Loan:** adds `Loan_type` (short / standard) and `Return_staff_id` (the staff member who checked the item in).
4. **Hold:** adds `Reserved_copy_id` (the specific copy set aside for a Ready hold).
5. **Fine:** a loan gets at most one fine of each type.
6. **Member:** adds an optional link to Staff, so an employee who also borrows is one person with two records.
7. **Mini-world wording clarified:** "periodicals" and "equipment" removed; "media" is defined; staff-borrower approval rule added.
8. **Replacement cost** stays nullable in the database, but the application requires a value for every item. Normal newspapers are priced like books (an individual cost per item); historical newspapers must be priced between $150 and $425, and research papers between $174 and $425. Newspapers gain an `Is_historical` flag so the two can be told apart.
9. **Catalog validation.** `Title` cannot be blank, and `Publication_year`, when given, must be between 1400 and 2100. The database cannot check "not in the future" because that depends on today's date, so the application enforces it.

Mini-World

We are planning to model a college/university library focused on the physical circulation of print materials and media. The library is managed by Staff employees who operate the system. Members register on the library's page, and they are of these types:

- Students, who can loan up to 99 items in total, up to 5 media items, and have no renewal limit on regular loans.
- Faculty, who can loan up to 400 items in total, up to 10 media items, and have no renewal limit.
- Staff (library employees borrowing as members), who can loan up to 199 items in total and 10 media items. **(v3)** An employee is both a Staff employee and a member of type Staff; the two records are linked.
- Alumni, who can loan up to 10 items in total, 4 media items, and renewals are capped at 1.

There is a difference between the permissions a Staff employee and a regular member have. Members can search the catalog, place and cancel their own holds, view their own loans and fines, pay a fine, and request a renewal depending on their member type.

Staff employees have all member-level access plus the following actions that members cannot perform:

- Check items in and out
- Register new members and edit member records, including deactivating an account
- Add, edit, or withdraw catalog items and copies
- Waive or adjust a fine amount
- View aggregate reports

Creating a loan record requires a Staff employee; a regular member cannot check an item out to themselves. They may only place a hold, which Staff will convert into a loan at pickup. **(v3)** When the borrower is a Staff employee, the

checkout (including picking up a hold) must be processed by a *different* Staff employee, so nobody approves their own loan.

Catalog

The catalog is based mostly on books, newspapers, and research papers, but also on audiovisual and electronic media such as movies, music, videogames, and board games. **(v3)**

Each catalog item belongs to exactly one category — Book, Newspaper, Research Paper, Videogame, Board Game, DVD, or Music — and is stored using a supertype/subtype design. Attributes common to every item (title, publisher, publication year, replacement cost) live on the general CatalogItem entity, while attributes specific to a category (such as an ISBN for a book, or a platform for a videogame) live on that category's own table. A catalog item's Item_id is shared between CatalogItem and its matching subtype row. **(v3)** The database enforces that the subtype row matches the item's category: a Book row cannot be attached to a videogame item.

Media vs. non-media (v3). Videogames, DVDs, music, and board games count as media (they count toward the member type's media-loan limit). Books, newspapers, and research papers do not. Every item counts toward the member type's total-loan limit.

Newspapers (v3). Each issue (a given newspaper on a given date and edition) is its own catalog item with its own copies, so a member can borrow a specific day or event. Newspapers circulate like any other item. A newspaper issue can be marked as **historical** (`Is_historical`), which changes its replacement-cost range (see the Fine System).

Research papers (v3). A research paper is a printed paper held by the library, with one or more authors. It circulates like any other item.

These items can be borrowed from the library using the loan system depending on the copies and availability of the item.

- If there is only 1 copy currently available of an item, a **short loan** applies: the member can borrow it for a maximum of 3 days, and renewal is not allowed.
- If there are 2 or more copies available of the item, a **standard loan** applies: it can be borrowed for up to 14 days, and renewals depend on the member type.

(v3) The loan records which kind of loan it is (`Loan_type`). A short loan must have 0 renewals and a due date no more than 3 days after checkout; a standard loan's due date is no more than 14 days after checkout, plus another 14 days for each renewal (a renewal extends the due date, so the limit has to grow with it).

A renewal extends the current due date without requiring the item to be returned first, but it is not automatic. A member may renew a standard loan up to the allowed limit, unless another member currently has an active or ready hold on that item. In that case, the renewal is denied so the item can be returned and passed along to the next member waiting for it. Since short loans already do not allow renewal, this check only applies to standard loans.

Hold System

If someone wants to borrow an item that is currently unavailable (all copies on loan), a hold system takes effect. A hold is placed at the item level, not on a specific copy, since any available copy of that item satisfies the request. Holds are served in order of hold_date. When a copy is checked in, the system finds the oldest active hold on that item and reserves that copy for the corresponding member. **(v3)** The reserved copy is recorded on the hold (`Reserved_copy_id`), and a Ready hold must have one. A member can have only one open hold (Active or Ready) per item at a time. Statuses:

- Active: waiting for a copy to become available
- Ready: a copy has been reserved for this member and is awaiting pickup
- Expired: the member did not pick up the item within the pickup window, and the hold releases to the next member in line
- Cancelled: the member cancelled the request
- Completed: the item was picked up and became a loan

A hold in "Ready" status expires if not picked up within 2 days, releasing the copy to the next hold in the queue.

Fine System

A fine is generated when a copy is returned after the agreed due date, when it remains overdue and unreturned, or when the returned copy is marked damaged or lost. It is tied to the specific loan that incurred it, not directly to the member. **(v3)** A loan can have at most one fine of each type (overdue, damaged, lost).

A fine has two statuses:

- Unpaid: assessed but not yet settled
- Paid: settled, with the date it was paid

Overdue fines: the amount is the days overdue times the daily rate ($0.35), capped at $10 per item. A fine is only created if the return date is after the due date, or if an item is still outstanding past the due date.

Damage/loss fines: rather than increasing by day, the amount equals the catalog item's replacement cost, since items vary significantly in value even within the same type. **(v3)** `Replacement_cost` is optional in the database, but the application requires a value for every item. Normal newspapers are priced individually, the same way as books. Historical newspapers must have a replacement cost between $150 and $425, and research papers between $174 and $425. A single loan may incur both an overdue fine and a damage/loss fine if applicable, such as when it is returned both late and damaged.

Staff can waive or adjust a fine by changing its amount (a waived fine is set to 0). A separate audit trail of waivers is outside the scope of this checkpoint.

A member with unpaid fines above $2.50 may be blocked from new checkouts or renewals.

Notification System

The system generates notifications to keep members informed about activity on their account. A notification is created when a hold becomes ready for pickup, when a loan is approaching its due date, when a loan becomes overdue, or when a fine is assessed. Each notification belongs to a single member and can be marked as read once viewed.

Notifications may be delivered through more than one channel: in-app (shown on the member's account page), email (sent to the member's email address, already stored on the Member entity), or postal mail (sent to the member's address, also already stored on Member). The same event may generate more than one notification record if it is delivered through more than one channel.

Entities, Attributes and Relationships

(20 tables. Rows marked **(v3)** are new or changed.)

1. MemberType — classifies members and sets their borrowing limits

Attribute	Type	Key	Constraints
Member_type_id	INT AUTO_INCREMENT	PK	NOT NULL
Type_name	VARCHAR(20)		NOT NULL, UNIQUE, not blank
Max_loans	INT		NOT NULL, CHECK > 0
Max_media_loans	INT		NOT NULL, CHECK > 0, CHECK <= Max_loans
Max_renewals	INT		NOT NULL, CHECK >= 0 (999 = unlimited)

2. Member — a person registered to borrow

Attribute	Type	Key	Constraints
Member_id	INT AUTO_INCREMENT	PK	NOT NULL
Member_type_id	INT	FK → MemberType	NOT NULL
Staff_id **(v3)**	INT	FK → Staff	nullable, UNIQUE (set only for employees who also borrow)
Username	VARCHAR(30)		NOT NULL, UNIQUE, not blank
First_name	VARCHAR(50)		NOT NULL, not blank
Last_name	VARCHAR(50)		NOT NULL, not blank
Email	VARCHAR(100)		NOT NULL, UNIQUE, valid email format
Phone	VARCHAR(20)		nullable, digits and phone symbols only, at least one digit
Street	VARCHAR(100)		nullable
City	VARCHAR(50)		nullable
State	VARCHAR(2)		nullable, two uppercase letters
Zip_code	VARCHAR(10)		nullable, 12345 or 12345-6789
Password_hash	VARCHAR(255)		NOT NULL, not blank
Date_registered	DATE		NOT NULL, default current date
Is_active	BOOLEAN		NOT NULL, default true

3. Staff — an employee who operates checkout, check-in, and member records

Attribute	Type	Key	Constraints
Staff_id	INT AUTO_INCREMENT	PK	NOT NULL
First_name	VARCHAR(50)		NOT NULL, not blank
Last_name	VARCHAR(50)		NOT NULL, not blank
Email	VARCHAR(100)		NOT NULL, UNIQUE, valid email format
Password_hash	VARCHAR(255)		NOT NULL, not blank
Is_active	BOOLEAN		NOT NULL, default true

4. CatalogItem — the supertype: attributes common to every catalog item

Attribute	Type	Key	Constraints
Item_id	INT AUTO_INCREMENT	PK	NOT NULL
Item_type **(v3)**	ENUM('book','videogame','boardgame','dvd','music','newspaper','researchpaper')		NOT NULL
Title	VARCHAR(255)		NOT NULL, not blank **(v3)**
Publisher_id	INT	FK → Publisher	nullable
Publication_year	SMALLINT		nullable, CHECK between 1400 and 2100 **(v3)** ("not in the future" is checked by the application)
Replacement_cost	NUMERIC(6,2)		nullable, CHECK >= 0
(Item_id, Item_type) **(v3)**		UNIQUE	lets every subtype link on id and type together

5. Publisher

Attribute	Type	Key	Constraints
Publisher_id	INT AUTO_INCREMENT	PK	NOT NULL
Name	VARCHAR(100)		NOT NULL, UNIQUE

6. Author

Attribute	Type	Key	Constraints
Author_id	INT AUTO_INCREMENT	PK	NOT NULL
First_name	VARCHAR(50)		NOT NULL
Last_name	VARCHAR(50)		NOT NULL

Subtype tables (v3 pattern). Every subtype has `Item_id` as its primary key plus an `Item_type` column that is fixed to its own value (default and CHECK). **Important:** every subtype's `Item_type` list must be written with the **same values in the same order** as CatalogItem (`'book','videogame','boardgame','dvd','music','newspaper','researchpaper'`). MySQL accepts a different order without an error, but then links rows by the ENUM's internal position instead of its text, so valid rows get rejected and wrong ones can get through. The foreign key links to CatalogItem on **(Item_id, Item_type)** together, so a subtype row can only attach to an item of the matching category. Tables 7, 9, 10, 11, 12, 13, and 14 all follow this pattern; the `Item_type` row is shown once per table below.

7. Book — subtype of CatalogItem

Attribute	Type	Key	Constraints
Item_id	INT	PK, FK → CatalogItem (with Item_type)	NOT NULL
Item_type **(v3)**	ENUM (same list as CatalogItem)	FK → CatalogItem (with Item_id)	NOT NULL, default 'book', CHECK = 'book'
Isbn	VARCHAR(13)		UNIQUE, nullable, 13 digits or 10 characters (digits, last may be X)
Genre	VARCHAR(50)		nullable

8. BookAuthor — join table, scoped to Book only

Attribute	Type	Key	Constraints
Item_id	INT	PK (composite), FK → Book	NOT NULL
Author_id	INT	PK (composite), FK → Author	NOT NULL

9. Videogame — subtype of CatalogItem

Attribute	Type	Key	Constraints
Item_id	INT	PK, FK → CatalogItem (with Item_type)	NOT NULL
Item_type **(v3)**	ENUM (same list)	FK → CatalogItem	NOT NULL, default 'videogame', CHECK = 'videogame'
Platform	VARCHAR(30)		nullable
Developer	VARCHAR(100)		nullable
Esrb_rating	VARCHAR(10)		nullable

10. BoardGame — subtype of CatalogItem

Attribute	Type	Key	Constraints
Item_id	INT	PK, FK → CatalogItem (with Item_type)	NOT NULL
Item_type **(v3)**	ENUM (same list)	FK → CatalogItem	NOT NULL, default 'boardgame', CHECK = 'boardgame'
Min_players	SMALLINT		nullable, CHECK > 0
Max_players	SMALLINT		nullable, CHECK > 0, CHECK >= Min_players
Designer	VARCHAR(100)		nullable

11. DVD — subtype of CatalogItem

Attribute	Type	Key	Constraints
Item_id	INT	PK, FK → CatalogItem (with Item_type)	NOT NULL
Item_type **(v3)**	ENUM (same list)	FK → CatalogItem	NOT NULL, default 'dvd', CHECK = 'dvd'
Runtime_minutes	SMALLINT		nullable
Director	VARCHAR(100)		nullable
Mpaa_rating	VARCHAR(10)		nullable

12. Music — subtype of CatalogItem

Attribute	Type	Key	Constraints
Item_id	INT	PK, FK → CatalogItem (with Item_type)	NOT NULL
Item_type **(v3)**	ENUM (same list)	FK → CatalogItem	NOT NULL, default 'music', CHECK = 'music'
Artist	VARCHAR(100)		nullable
Track_count	SMALLINT		nullable

13. Newspaper (v3) — subtype of CatalogItem; one row per issue

Attribute	Type	Key	Constraints
Item_id	INT	PK, FK → CatalogItem (with Item_type)	NOT NULL
Item_type	ENUM (same list)	FK → CatalogItem	NOT NULL, default 'newspaper', CHECK = 'newspaper'
Newspaper_name	VARCHAR(100)		NOT NULL
Issue_date	DATE		NOT NULL, indexed (members search by date)
Edition	VARCHAR(30)		NOT NULL, default '' (empty = single edition)
Is_historical	BOOLEAN		NOT NULL, default false; true = historical issue (replacement cost 150–425)
(Newspaper_name, Issue_date, Edition)		UNIQUE	the same issue cannot be entered twice

14. ResearchPaper (v3) — subtype of CatalogItem

Attribute	Type	Key	Constraints
Item_id	INT	PK, FK → CatalogItem (with Item_type)	NOT NULL
Item_type	ENUM (same list)	FK → CatalogItem	NOT NULL, default 'researchpaper', CHECK = 'researchpaper'
Doi	VARCHAR(100)		UNIQUE, nullable
Journal_name	VARCHAR(150)		nullable
Volume	VARCHAR(20)		nullable
Issue	VARCHAR(20)		nullable
Page_range	VARCHAR(20)		nullable

15. PaperAuthor (v3) — join table, scoped to ResearchPaper only

Attribute	Type	Key	Constraints
Item_id	INT	PK (composite), FK → ResearchPaper	NOT NULL
Author_id	INT	PK (composite), FK → Author	NOT NULL

16. Copy — one physical instance of a catalog item

Attribute	Type	Key	Constraints
Copy_id	INT AUTO_INCREMENT	PK	NOT NULL
Item_id	INT	FK → CatalogItem	NOT NULL
Barcode	VARCHAR(20)		NOT NULL, UNIQUE
Condition	ENUM('new','good','worn','damaged')		NOT NULL
Shelf_location	VARCHAR(30)		nullable
Status	ENUM('available','on loan','lost','withdrawn')		NOT NULL, default 'available'
Date_acquired	DATE		NOT NULL

17. Loan

Attribute	Type	Key	Constraints
Loan_id	INT AUTO_INCREMENT	PK	NOT NULL
Copy_id	INT	FK → Copy	NOT NULL
Member_id	INT	FK → Member	NOT NULL
Staff_id	INT	FK → Staff	NOT NULL (staff who checked the item out)
Return_staff_id **(v3)**	INT	FK → Staff	nullable (staff who checked it in); filled exactly when Return_date is filled
Loan_type **(v3)**	ENUM('short','standard')		NOT NULL; short: Renewal_count = 0 and due within 3 days; standard: due within 14 days plus 14 per renewal
Checkout_date	TIMESTAMP		NOT NULL, default current timestamp
Due_date	DATE		NOT NULL, CHECK after Checkout_date (at least the next day)
Return_date	DATE		nullable, CHECK not before the checkout day
Renewal_count	SMALLINT		NOT NULL, default 0, CHECK >= 0
Open_copy_id	INT (generated)	UNIQUE	equals Copy_id while the loan is unreturned, else NULL; guarantees at most one open loan per copy

18. Hold

Attribute	Type	Key	Constraints
Hold_id	INT AUTO_INCREMENT	PK	NOT NULL
Item_id	INT	FK → CatalogItem	NOT NULL
Member_id	INT	FK → Member	NOT NULL
Reserved_copy_id **(v3)**	INT	FK → Copy	nullable; required when Status = 'Ready'
Hold_date	TIMESTAMP		NOT NULL, default current timestamp
Status	ENUM('Active','Ready','Expired','Cancelled','Completed')		NOT NULL, default 'Active'
Expiry_date	DATE		nullable, not before the hold day
Is_open	TINYINT (generated)		1 while Active or Ready, else NULL; with (Item_id, Member_id) it is UNIQUE, so a member has at most one open hold per item

19. Fine

Attribute	Type	Key	Constraints
Fine_id	INT AUTO_INCREMENT	PK	NOT NULL
Loan_id	INT	FK → Loan	NOT NULL
Fine_type	ENUM('overdue','damaged','lost')		NOT NULL
Amount	NUMERIC(6,2)		NOT NULL, CHECK >= 0
Date_assessed	DATE		NOT NULL, default current date
Date_paid	DATE		nullable, not before Date_assessed
Is_paid	BOOLEAN		NOT NULL, default false; true exactly when Date_paid is filled
(Loan_id, Fine_type) **(v3)**		UNIQUE	at most one fine of each type per loan

20. Notification

Attribute	Type	Key	Constraints
Notification_id	INT AUTO_INCREMENT	PK	NOT NULL
Member_id	INT	FK → Member	NOT NULL
Notification_type	ENUM('hold_ready','due_soon','overdue','fine_assessed')		NOT NULL
Channel	ENUM('in_app','email','postal')		NOT NULL, default 'in_app'
Message	VARCHAR(255)		NOT NULL
Date_sent	TIMESTAMP		NOT NULL, default current timestamp
Is_read	BOOLEAN		NOT NULL, default false

Relationships

- A member type classifies many members; a member has exactly one member type (1:N)
- A member may have many loans; a loan belongs to exactly one member (1:N)
- A staff member processes many loans at checkout, and may process many returns; each loan has exactly one checkout staff member and at most one check-in staff member (1:N, twice) **(v3)**
- A staff employee may also be a member; the optional link is 1:1 **(v3)**
- A publisher publishes many catalog items; a catalog item has at most one publisher (1:N)
- A catalog item is specialized into exactly one subtype — Book, Newspaper, ResearchPaper, Videogame,

BoardGame, DVD, or Music — determined by its Item_type; the subtype's Item_id is both its primary key and, together with Item_type, a foreign key back to CatalogItem (1:1) **(v3)**
- A book may have many authors, and an author may write many books (M:N via BookAuthor)
- A research paper may have many authors, and an author may write many papers (M:N via PaperAuthor) **(v3)**
- A catalog item has many copies; a copy belongs to exactly one catalog item (1:N)
- A copy may be lent through many loans over its lifetime but is on at most one open loan at a time (1:N)
- A member may place many holds; a hold belongs to exactly one member (1:N)
- A catalog item may have many holds against it; a hold refers to exactly one catalog item (1:N)
- A Ready hold refers to one specific reserved copy of that item (N:1) **(v3)**
- A loan may incur more than one fine, but at most one of each type (1:N)
- A member may have many notifications; a notification belongs to exactly one member (1:N)

Rules Enforced by the Application (not by table constraints)

These compare data across tables or depend on the current moment, so the application (or a later trigger) handles them:

- A member cannot exceed their member type's total-loan or media-loan limit, and cannot renew more than Max_renewals.
- Short vs. standard loan is chosen from the number of copies available at checkout; the renewal-denied-when-another-hold-exists rule.
- A checkout for a Staff-employee borrower must be processed by a different Staff employee.
- Overdue and damage/loss fine amounts, and blocking checkouts for unpaid fines above $2.50.
- Keeping Copy.Status in step with loans (on loan / available) and releasing expired Ready holds.
- Inactive members cannot borrow.
- Publication year cannot be in the future (the database only enforces the 1400–2100 range).
- Replacement cost is required for every item. Ranges: historical newspapers 150–425, research papers 174–425; normal newspapers are priced like books. (These compare a value in CatalogItem with a flag or type in another table, so the application enforces them.)

Open Items

- Capitalization of table and column names (for example `Videogame` vs `VideoGame`) — follow this document exactly.